

Vivek Kumar

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Advanced Application Engineer specializing in orchestrating end-to-end domain-specific software platforms powered by Gen AI and agentic codebases. Proven track record of designing cutting-edge product prototypes, scalable cloud-native architectures, and deploying IP-led solutions in Hyper Agile environments.

EDUCATION

- Birla Institute of Technology, Mesra** 2022 – 2027
Integrated M.Sc in Quantitative Economics and Data Science CGPA: 7.05

RESEARCH & PUBLICATIONS

- Vivek Kumar**, Dr. Mrinal Jana. “Distribution-free Uncertainty Quantification for Option Pricing Using Residual-space Conformal Prediction.” *CIMA 2026, Accepted*. Proceedings to appear in Springer *LNNS* (Paper ID-498, Scopus-indexed).
Developed a novel mathematical framework for quantifying algorithmic uncertainty, directly applicable to risk mitigation and reliability testing in complex, high-stakes application environments.

APPLICATION ENGINEERING & GEN AI PLATFORMS

- Agentic Memory System for LLMs (Advanced RAG) | Neurohack (26th Rank)** *FastAPI, SQLite, NLP, Embeddings*
 - Secured 26th rank nationally (Certificate) by architecting a scalable, stateful memory system for LLMs handling 1,000+ conversation turns, establishing a core component for **agentic codebases**.
 - Engineered a production-ready, hybrid Retrieval-Augmented Generation (RAG) pipeline combining semantic embeddings with lexical BM25 ranking, achieving sub-100ms retrieval latency via highly optimized FastAPI endpoints.
- End-to-End Gen AI & Analytics Platform** *Python, SQL, Streamlit, Gen AI, Docker, AWS*
 - Orchestrated an end-to-end domain-specific software platform in a **Hyper Agile environment**, translating complex client logic into a scalable, data-driven application prototype.
 - Built an innovative product prototype with Gemini Gen AI integration, empowering non-technical stakeholders to execute natural-language queries against real-time data to drive **IP-led services**.
 - Designed and deployed a fully automated data pipeline ingesting from 5+ heterogeneous live sources daily, applying strict data validation and monitoring protocols across 126+ tracked entities.
 - Operationalized the solution using cloud-native microservices with Docker, Git-based CI/CD (GitHub Actions), and AWS EC2 for reproducible, automated execution.
- Heuristic Optimization Engine & Scenario Analysis** *Python, SciPy, Pandas, NumPy*
 - Designed and developed a cutting-edge algorithmic prototype utilizing a Regime-Aware Genetic Algorithm from scratch to solve high-dimensional optimization problems.
 - Orchestrated rigorous validation testing via 10,000 bootstrap resamples, proving the architecture’s structural outperformance over traditional benchmarks in 92% of synthetic scenarios.
- Advanced Statistical Prototyping (Metropolis-Hastings)** *R, Probabilistic Programming*
 - Designed complex statistical prototypes using Metropolis-Hastings MCMC (40,000 iterations) to orchestrate high-dimensional probabilistic inference.
 - Evaluated complex algorithmic solutions using rigorous trace diagnostics and sensitivity analysis; documented architectural assumptions and data-quality limitations for technical stakeholders.

TECHNICAL SKILLS

Gen AI & Agentic Systems: Large Language Models (LLMs), Advanced RAG, Multi-Agent Workflows, Context Window Management, Semantic Search, Prompt Engineering
Application Engineering: Product Prototyping, Scalable APIs (FastAPI), End-to-End Orchestration, Microservices, Hyper Agile Methodologies
Cloud Native & DevOps: Docker, CI/CD (GitHub Actions), AWS EC2, Git, Vector Similarity Search
Languages & Data: Python, SQL, C++, R, Pandas, NumPy, ETL Pipelines, SQLite
Algorithms: Optimization Algorithms, Statistical Inference, Predictive Modeling, A/B Testing

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

- Events & Public Relations Organizer | Data Science Summit '23, BITOTSAV '24**
 - Spearheaded corporate outreach for the Data Science Summit, successfully pitching tech sponsors and coordinating cross-functional Agile teams to deliver successful large-scale events.